

HIRANYA AGARWAL

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EDUCATION

Bachelor of Engineering (Honours), Mechanical Engineering | UNSW Sydney 2025 – 2028 (expected)
SACE | Pedare Christian College, Golden Grove SA — ATAR: 97.05 2024

TECHNICAL SKILLS

CAD & mechanical design: SolidWorks, Fusion 360 — drone airframes, competition robots, antenna mounts and brackets
Programming: C++ (embedded & competition robotics), Python (automation, data analysis, trading APIs), MATLAB, TypeScript
Mechatronics & embedded: Sensor integration (IMU, encoders, accelerometers, colour), flight controllers, pneumatics, Raspberry Pi & ADC data acquisition

EXPERIENCE

Mechanical Lead | BlueSat UNSW — CubeSat Program Feb 2025 – Present
– Lead mechanical design of the ground-station antenna system for a student-built CubeSat, iterating mount and bracket designs for the antenna pointing assembly.
– Coordinate integration across electrical, software and mechanical subteams toward launch milestones, and help run outreach events promoting space engineering at UNSW.

VEX Robotics Mentor (Casual) | Barker College, Hornsby NSW 2026 – Present
– Coach VEX IQ teams through design, build and programming cycles, drawing on national-championship VRC competition experience.

Co-Founder | Project Umbrella — UNSW hands-on engineering society 2025 – Present
– Co-founded a student society running practical, build-focused engineering projects and workshops for undergraduates.

Front of House | Caffe Primo, Munno Para SA Dec 2021 – Jan 2025
– Handled orders, payments and bar service in a high-volume hospitality environment.

PROJECTS

ADA²M — Autonomous Agricultural Monitoring Drone | Personal project, team of 2 2025 – Present
– Building a fully autonomous drone that sweeps pre-defined flight paths over farmland collecting real-time environmental data (temperature, humidity, air quality) for small-scale farmers.
– SpeedyBee F405 V4 flight controller, TBS Crossfire long-range RC link and a custom carbon-fibre frame designed in SolidWorks; Jetson Nano runs onboard inference for flight decisions. Currently in flight testing.

Cafe Primo Firle - Restaurant Operations Platform | Working restaurant 2026 - Present
– Built a production restaurant platform connecting customer ordering, Stripe checkout, live staff and admin dashboards, PostgreSQL, real-time Socket.io updates and ESC/POS receipt printing.

Automated Day-Trading Platform | Personal project 2025 – Present
– Python trading system on the Interactive Brokers API with modular backtesting, position sizing, stop-loss logic and drawdown limits validated against historical market data.

LEADERSHIP & VOLUNTEERING

Captain & Lead Programmer | VEX Robotics Team 41103A, Pedare Christian College 2018 – 2024
– Led a 7-person team to back-to-back Australian National Championships and two VEX World Championship qualifications; Sportsmanship Award at Worlds 2024 (Dallas, TX).
– Wrote all autonomous and driver-control code in C++, fusing IMU, colour-sensor and encoder data for precise field positioning; designed pneumatic intake, scoring and elevation mechanisms.
– Mentored junior teams (2020–2023) and helped organise the 2022 Australian National VEX Championship.

Volunteer | Little Sisters of the Poor (weekly) & Order of Malta street outreach, Sydney Feb 2024 – Present
– Weekly maintenance and companionship visits with elderly residents; distribute essential supplies to people experiencing homelessness through organised street outreach.

AWARDS

Australian Defence Force Future Innovators Award (2024) • Tournament Excellence, Australian National VEX Championship (2023) • Robotics work featured on ABC Radio, LifeFM Adelaide and in Glam Adelaide (2023)